

Patient name: Jennifer Pei Ru Hao
DOB: February 15, 1994
Sex: Female

Kit ID: TSB258
System version: 4.5.2

Sample Collected: August 20, 2025
Sequence Sample Received: August 27, 2025
Sequence Results Ready: September 16, 2025
Report Generated: September 16, 2025

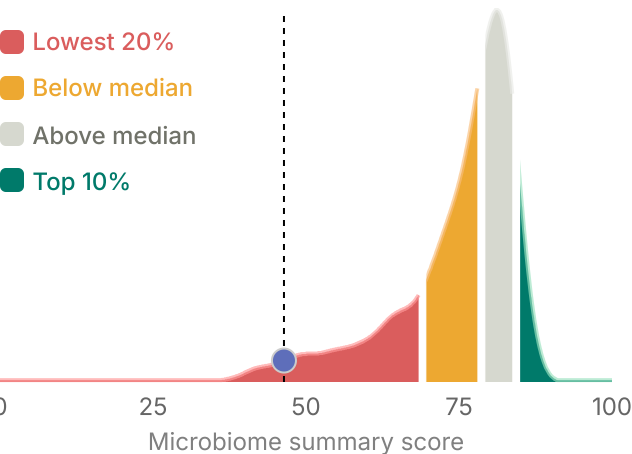
Microbiome Summary Score

Your microbiome score evaluates all the relevant metrics for your age range into a single number, giving you a clear picture of how your microbiome stacks up. The score ranges from 0 to 100. A score closer to 100 suggests your microbiome is in excellent shape compared to the reference population. Conversely, a score closer to 0 indicates more areas for improvement.

We calculate this score by looking at various factors in your gut/vaginal microbiome, such as the balance of beneficial bacteria versus opportunistic pathogens. We then compare these factors to a wide range of individuals. Our dataset includes publicly available reference samples and proprietary data from Tiny Health customers.

To determine your percentile, we compare your Microbiome Summary Score to others in the same age range to help you understand what is typical and where your sample falls. For example, if you are in the 41st percentile, 41% of comparable samples have a lower Microbiome Summary Score and 59% have a higher score. PRO and Beta metrics are not included in MSS calculations. MSS for pregnant individuals is calculated separately and should only be compared to other pregnant samples.

46 Microbiome summary score



Sample at a glance 21 Needs support 11 Needs improvement

Metabolic Health ⚠ Akkermansia 0.000%	⚠ Citrobacter 9.624% ⚠ Enterobacter 0.092%	Microbiome Diversity ⚠ Shannon diversity 4.44
Anti-inflammatory Markers ⚠ Faecalibacterium 0.000%	Mucus Degradation Index ⚠ Mucus degradation index 17	Overabundant Species ⚠ Bacteroides fragilis 3.540% ⚠ Phocaeicola dorei 16.267%
Antibiotic Resistance Signature ⚠ Abundance index 0.28 ⚠ Richness index 0.42	Hexa-LPS Index ⚠ Hexa-LPS index 16	Gut Resilience Score ⚠ Gut resilience score 0.39
Opportunistic Pathogens ⚠ Enterobacteriaceae 20.673% ⚠ Klebsiella 0.300% ⚠ Klebsiella pneumoniae 0.150% ⚠ Klebsiella oxytoca 0.137% ⚠ Escherichia coli 6.694% ⚠ Escherichia flexneri 3.610% ⚠ Escherichia dysenteriae 0.096%	Hydrogen Sulfide Index ⚠ Hydrogen sulfide index 17	Major Bacterial Phyla ⚠ Firmicutes 22.044% ⚠ Proteobacteria 20.701%
	Fiber Digestion Capacity ⚠ Resistant starch 2465 rpk	Gut Ratio ⚠ Proteobacteria / Actinobacteriota ratio 14
	Protein Breakdown Capacity ⚠ Protein breakdown 790 rpk ⚠ Trimethylamine 114 rpk ⚠ Ammonia 4103 rpk	Histamine-producing Species ⚠ Histamine-producing species 0.010%

Results Summary

	Metrics	Clinical indication examples
Beneficial microbes	✔ Beneficial Bifidobacterium ⚠ Metabolic Health ⚠ Anti-inflammatory Markers ✔ Common Probiotic Species	• Prebiotics / probiotics support • Review Akkermansia promoting supplements • Track and diversify fiber intake • Increase fermented foods • Move and stay physically active

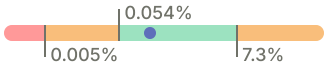
Balance and robustness	⚠️ Microbiome Diversity ⚠️ Overabundant Species ⚠️ Gut Resilience Score ✅ Oral Microbes in the Gut ⚠️ Major Bacterial Phyla ✅ Common Microbiome Members ⚠️ Gut Ratio - Track and diversify fiber intake - Increase fermented foods - Use non-toxic cleaners / reduce antimicrobials
Microbial enzymes & metabolites	⚠️ Histamine-producing Species ⚠️ Complex Compound Breakdown ⚠️ Microbial GABA Capacity ⚠️ Modified Bile Acid Production Capacity ✅ Mitochondrial Health Support

Detailed Gut Health Test Results

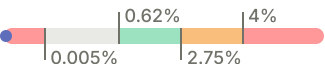
Legend: (p) Phylum (f) Family (g) Genus (sp) Species

Beneficial microbes

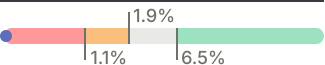
Beneficial Bifidobacterium

 Bifidobacterium (g)	0.395%	
Bifidobacterium animalis (sp)	0.122%	
Bifidobacterium breve (sp)	0.072%	
Bifidobacterium dentium (sp)	0.079%	
Bifidobacterium longum (sp)	0.119%	

Metabolic Health

 Akkermansia (g)	0.000%	
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Anti-inflammatory Markers

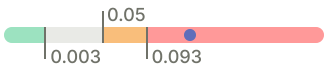
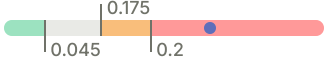
 Faecalibacterium (g)	0.000%	
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Common Probiotic Species

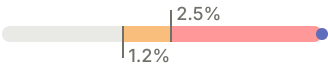
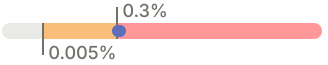
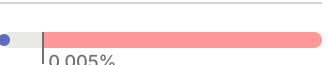
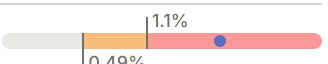
 Bifidobacterium (g)	0.395%	
 Lactobacillaceae (f)	0.011%	
 Bifidobacterium longum (sp)	0.119%	
 Bifidobacterium breve (sp)	0.072%	
 Bifidobacterium animalis (sp)	0.122%	

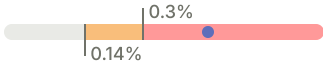
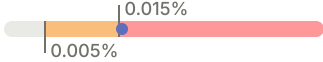
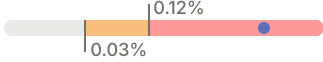
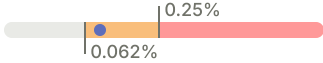
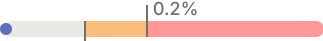
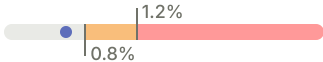
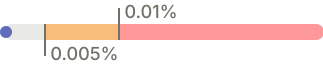
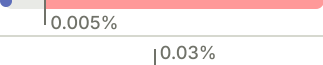
Disruptive microbes

Antibiotic Resistance Signature

 Abundance index	0.28	
 Richness index	0.42	

Opportunistic Pathogens

 Enterobacteriaceae (f)	20.673%	
Citrobacter freundii (sp)	0.224%	
Citrobacter portucalensis (sp)	3.232%	
Citrobacter werkmanii (sp)	0.268%	
Citrobacter_A amalonaticus (sp)	3.099%	
Citrobacter_A amalonaticus_C (sp)	0.057%	
Citrobacter_A farmeri (sp)	2.536%	
Citrobacter_A sp009363175 (sp)	0.167%	
Enterobacter hormaechei_A (sp)	0.080%	
Escherichia coli (sp)	0.103%	
Escherichia coli_D (sp)	6.561%	
Escherichia dysenteriae (sp)	0.096%	
Escherichia flexneri (sp)	3.610%	
Klebsiella pneumoniae (sp)	0.150%	
Klebsiella_A oxytoca (sp)	0.137%	
 Klebsiella (g)	0.300%	
Klebsiella pneumoniae (sp)	0.150%	
Klebsiella_A oxytoca (sp)	0.137%	
 Klebsiella pneumoniae (sp)	0.150%	
 Klebsiella oxytoca (sp)	0.137%	
 Salmonella enterica (sp)	0.000%	
 Escherichia coli (sp)	6.694%	

Escherichia coli (sp)	0.103%	
Escherichia coli_D (sp)	6.561%	
⚠ Escherichia flexneri (sp)	3.610%	
⚠ Escherichia dysenteriae (sp)	0.096%	
⚠ Citrobacter (g)	9.624%	
Citrobacter freundii (sp)	0.224%	
Citrobacter portucalensis (sp)	3.232%	
Citrobacter werkmanii (sp)	0.268%	
Citrobacter_A amalonaticus (sp)	3.099%	
Citrobacter_A amalonaticus_C (sp)	0.057%	
Citrobacter_A farmeri (sp)	2.536%	
Citrobacter_A sp009363175 (sp)	0.167%	
⚠ Enterobacter (g)	0.092%	
Enterobacter hormaechei_A (sp)	0.080%	
✔ Morganella (g)	0.000%	
✔ Raoultella (g)	0.000%	
✔ Streptococcus (g)	0.390%	
Streptococcus salivarius (sp)	0.145%	
Streptococcus sp001556435 (sp)	0.115%	
✔ Staphylococcus (g)	0.000%	
✔ Pseudomonas aeruginosa (sp)	0.000%	
✔ Haemophilus influenzae (sp)	0.000%	
✔ Haemophilus parainfluenzae (sp)	0.006%	

✓ Enterococcus faecium (sp)	0.000%	0.005%
✓ Enterococcus faecalis (sp)	0.000%	0.005%
✓ Clostridioides difficile (sp)	0.000%	0.005% 0.007%
✓ Clostridium perfringens (sp)	0.000%	0.05% 0.1%
✓ Acinetobacter baumannii (sp)	0.000%	0.005%
✓ Campylobacter (g)	0.000%	0.005%
Potential Stomach Inflammation		
✓ Helicobacter pylori (sp)	0.000%	0.005%
Parasites and Infection		
✓ Blastocystis (g)	0.000%	0.005%
✓ Cryptosporidium (g)	0.000%	0.005%
✓ Entamoeba histolytica (sp)	0.000%	0.005%
✓ Entamoeba dispar (sp)	0.000%	0.005%
✓ Giardia (g)	0.000%	0.005%
✓ Yersinia enterocolitica (sp)	0.000%	0.005%
✓ Vibrio (g)	0.000%	0.005%
✓ Vibrio cholerae (sp)	0.000%	0.005%
✓ Cyclospora cayetanensis (sp)	0.000%	
Potential Fungal Overgrowth		
✓ Candida (g)	0.000%	0.005%
✓ Aspergillus (g)	0.000%	0.005%
✓ Cryptococcus (g)	0.000%	0.005%
✓ Saccharomyces (g)	0.019%	0.5%
✓ Rhodotorula (g)	0.000%	0.005%
✓ Saprochaete (g)	0.000%	0.005%

✓ Malassezia (g)	0.000%	0.005%
✓ Microsporum (g)	0.000%	0.005%
✓ Trichophyton (g)	0.000%	0.005%
Methane Production		
✓ Methanobrevibacter smithii (sp)	0.000%	0.15% 0.44%
✓ Methane production capacity	0.00 rpkm	1

Gut inflammation markers

Mucus Degradation Index

⚠ Mucus degradation index	17.44	8.69 11.7 12.4
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Hexa-LPS Index

⚠ Hexa-LPS index	16.56	9.58 9.82
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Hydrogen Sulfide Index

⚠ Hydrogen sulfide index	17.08	9.12 10.7 11.4
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Host DNA

✓ Host DNA	0.080%	5%
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Short-chain fatty acids

Butyrate Capacity

✓ Butyrate	1214.50 rpkm	910 943 1230
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Propionate Capacity

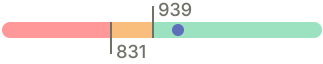
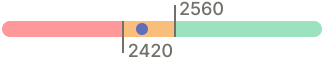
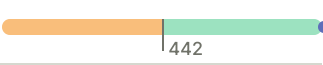
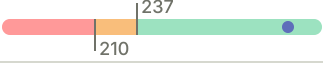
✓ Propionate	770.20 rpkm	533 547 706
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Acetate Capacity

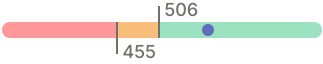
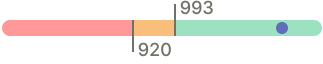
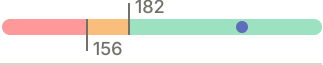
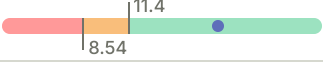
✓ Acetate	838.16 rpkm	700 719 920
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Digestion & absorption markers

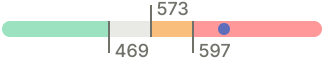
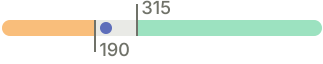
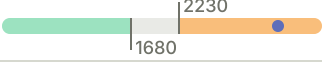
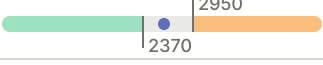
Fiber Digestion Capacity

✓ Cellulose	1216.66 rpkm	
⚠ Resistant starch	2465.77 rpkm	
✓ Chitin	1340.49 rpkm	
✓ Pectin	850.41 rpkm	

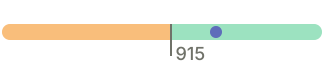
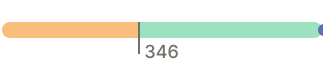
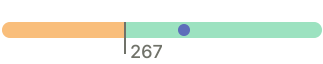
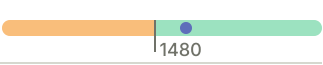
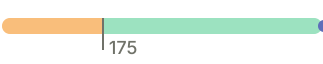
Complex Sugar Digestion Capacity

✓ Fructooligosaccharides	790.84 rpkm	
✓ Galactooligosaccharides	2065.16 rpkm	
✓ Xylooligosaccharides	566.20 rpkm	
✓ Isomaltooligosaccharides	96.84 rpkm	

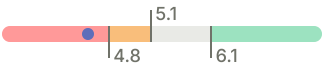
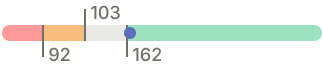
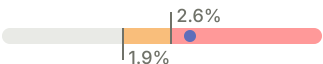
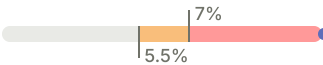
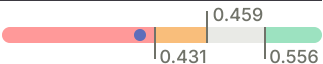
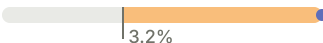
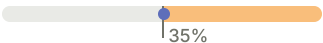
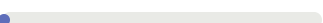
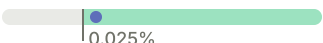
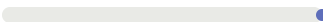
Protein Breakdown Capacity

⚠ Protein breakdown	790.77 rpkm	
✓ Indole-3-propionic acid	219.89 rpkm	
⚠ Trimethylamine	114.36 rpkm	
⚠ Ammonia	4103.35 rpkm	
✓ Branched Chain Amino Acids	2580.03 rpkm	

Vitamin Production Capacity

✓ Vitamin B2	1226.21 rpkm	
✓ Vitamin B7	1037.78 rpkm	
✓ Vitamin B9	443.57 rpkm	
✓ Vitamin B12	1858.56 rpkm	
✓ Vitamin K	840.89 rpkm	

Balance and robustness

Microbiome Diversity		
⚠ Shannon diversity	4.44	
✅ Species richness	164.00 species	
Overabundant Species		
⚠ Bacteroides fragilis (sp)	3.540%	
⚠ Phocaeicola dorei (sp)	16.267%	
Gut Resilience Score		
⚠ Gut resilience score	0.39	
Oral Microbes in the Gut		
✅ Oral microbes	0.137%	
Major Bacterial Phyla		
✅ Bacteroidota (p)	55.034%	
⚠ Firmicutes (p)	22.044%	
✅ Actinobacteriota (p)	1.477%	
⚠ Proteobacteria (p)	20.701%	
✅ Fusobacteriota (p)	0.000%	
Common Microbiome Members		
✅ Bacteroides (g)	34.798%	
✅ Bacteroides fragilis (sp)	3.540%	
✅ Prevotella (g)	0.114%	
✅ Ruminococcus (g)	0.010%	
✅ Ruminococcus gnavus (sp)	0.000%	
✅ Blautia (g)	0.088%	
✅ Roseburia (g)	0.006%	
✅ Phocaeicola dorei (sp)	16.267%	

Gut Ratio



Microbial enzymes & metabolites

Histamine-producing Species



Complex Compound Breakdown



Microbial GABA Capacity



Modified Bile Acid Production Capacity

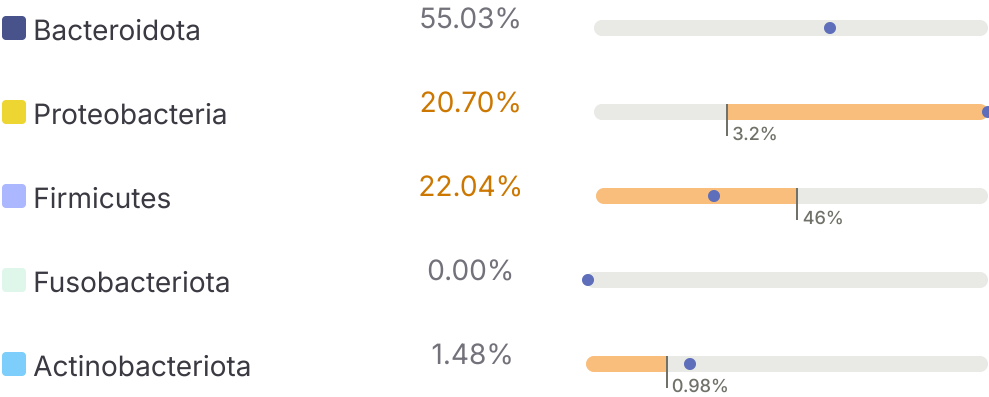


Mitochondrial Health Support

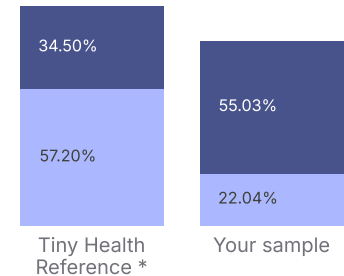


Major Microbial Members & Ratios

Common Bacterial Phyla



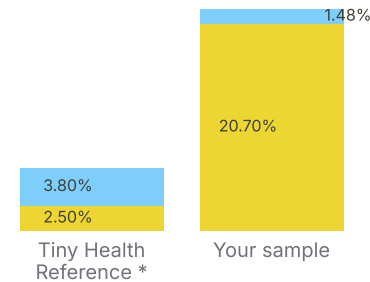
Gut Ratio



Firmicutes:Bacteroidota Ratio 0.40 (Ideally <30)

Firmicutes and Bacteroidota are the two largest phyla that make up the gut microbiome in humans. Historically and primarily in 16S studies, the ratio of Firmicutes/Bacteroidota has been associated with a number of conditions in adults, including obesity, Type 2 Diabetes, and inflammation. However, the use of these markers is now considered dated by current microbiome researchers, and the validity of these associations has been called into question.

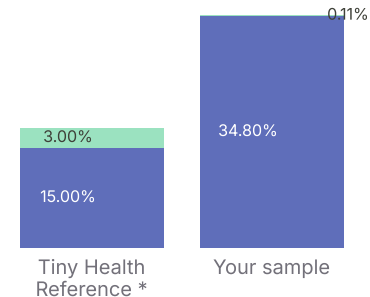
Firmicutes Bacteroidota



Proteobacteria:Actinobacteriota Ratio 14.02 (Ideally <14)

Aside from Firmicutes and Bacteroidota, Proteobacteria and Actinobacteriota are two common phyla in the gut microbiome in humans. Proteobacteria are widely associated with pathogens, while Actinobacteriota are associated with commensal species. Historically in primarily 16S studies, the ratio of Proteobacteria:Actinobacteriota has been associated with gut inflammation. However, the use of these markers is now considered dated by current microbiome researchers, and the validity of these associations has been called into question.

Proteobacteria Actinobacteriota



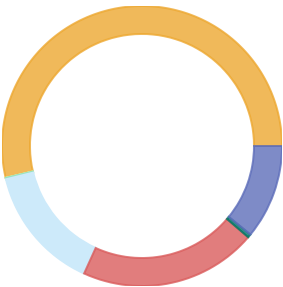
Prevotella:Bacteroides Ratio 0.00 (Ideally >4)

Prevotella and Bacteroides are two common genera in the gut microbiome in humans. One or the other often dominates the gut microbiome, and can define the Gut Type. Prevotella are widely associated with a high fiber diet, while Bacteroides are associated with a diet high in fat and protein. Historically in primarily 16S studies, the ratio of Prevotella:Bacteroides has been associated with body weight, fat loss, and diet success. However, the use of these markers is now considered dated by current microbiome researchers, and the validity of these associations has been called into question.

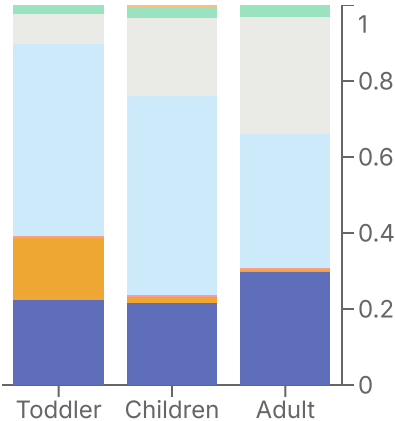
Bacteroides Prevotella

*Tiny Health Reference: This represents a cohort that is free of any acute or chronic conditions, so they may be considered a "healthy" cohort

Gut Type



Bacteroides	53.63%
Ruminococcaceae	0.01%
Prevotella	0.16%
Lachnospiraceae	14.39%
Enterobacteriaceae	20.48%
Bifidobacterium	0.40%
Other	10.90%



These bars represent the proportion of the different gut types in our database for different age ranges. The gut type of a sample is determined by the dominant type of microbe in that sample.

Type Bacteroides

The *Bacteroides* gut type is a common type for all ages. As it turns out, this is the most common gut type for people in developed countries. It's often seen in those whose diet is high in animal protein and fats, with lower consumption of plant-based foods.

When compared to the other gut types, the *Bacteroides* gut type has the highest potential to produce vitamin B12 and biotin.

Studies have shown that gut types respond differently to dietary interventions. For example, while foods high in fiber are beneficial for all gut types, if trying to lose weight, the *Bacteroides* gut type may not benefit much from a fiber-rich diet.

Compared to the *Prevotella* gut type, a good thing about the *Bacteroides* gut type is that it tends to have higher levels of *Bifidobacterium* species. In fact, babies with a gut type dominated by *Bifidobacterium* often transition to a *Bacteroides* gut type during toddlerhood.

Compared to other gut types, a *Bacteroides* gut type seems to benefit the most from eating chili peppers, whose content of capsaicin has beneficial effects on the gut microbiome and overall health.

Different gut types also respond differently to supplements. One study found that when people with a *Bacteroides* gut type were given probiotics, there was an improvement in glucose metabolism. Instead, people with a *Prevotella* gut type saw improvement in triglyceride metabolism.

Researchers have identified two subtypes in the *Bacteroides* gut type, named Bact1 and Bact2. The Bact2 subtype has three main characteristics in terms of microbiome composition:

- Top species include *Bacteroides fragilis*, *Phocaeicola vulgatus*, and *Parabacteroides distasonis*
- Has a low diversity
- Has very low levels of *Faecalibacterium prausnitzii*

The Bact2 subtype is more often seen in people with obesity, insulin resistance, or inflammatory conditions such as inflammatory bowel disease and multiple sclerosis.

Microbiome Breakdown

- **Beneficial:** These support microbiome health and overall health.
- **Unfriendly:** Low levels of these microbes are expected. Higher levels for extended periods of time can cause issues. This should make up a minority of the microbiome, kept in check by beneficial and variable microbes. Any unfriendly microbes out of range will be flagged in the "Detailed Gut Health Test Results" section above.
- **Variable:** These can be beneficial or unfriendly depending on their levels and what other microbes are around them. Lower levels are usually ok, and higher levels can cause issues.
- **Unknown:** These are lesser known microbes. Not enough is known to classify them as beneficial, unfriendly or variable at this time.

Your breakdown



Top 20 species

Metric	Result	Description
● Phocaeicola dorei	16.267%	<i>P. dorei</i> are very common in the gut of adults and babies. Some studies have found that in children with a family history of type 1 diabetes (T1D), high levels of <i>P. dorei</i> may be associated with T1D development. In babies, it's been found that the risk of developing T1D increases when <i>P. dorei</i> levels are high, <i>Bifidobacterium</i> are low, and <i>Enterobacteriaceae</i> are absent to train the baby's immune system. When present at levels higher than ideal, this will be flagged in the Balance and Robustness category of your results.
● Bacteroides thetaiotaomicron	10.575%	<i>B. thetaiotaomicron</i> is a common gut member in healthy people. In babies, it digests breastmilk sugars (HMOs) and contributes to the production of the beneficial molecule butyrate. It can suppress the production of toxins by unfriendly gut bacteria, like <i>Clostridioides difficile</i> and <i>Escherichia coli</i>. Due to its anti-inflammatory properties it's being studied as a biotherapeutic for Crohn's disease. However, like other <i>Bacteroides</i> species, it can become infectious outside of the gut if the opportunity presents.

● Bacteroides caccae	9.956%	<i>B. caccae</i> is a variable bacterium in your gut microbiome. Laboratory experiments suggest that it contributes to gut health by protecting the gut barrier from unfriendly bacteria. However, if your <i>B. caccae</i> levels are too high, you may get into trouble. High amounts of <i>B. caccae</i> may contribute to inflammation and development of: • Type 2 diabetes • Inflammatory bowel disease • Obesity Rarely, it can cause serious blood infections. Luckily, you can keep your levels of <i>B. caccae</i> in the sweet spot by eating a healthy anti-inflammatory diet.
● Bacteroides ovatus	7.262%	Lots of healthy people have <i>B. ovatus</i> in their gut. This bacterium is good at digesting plant-based foods. It contributes to gut health by modulating the immune response through its anti-inflammatory properties. However, like other <i>Bacteroides</i> species, it can become infectious outside of the gut if the opportunity presents. Also, high numbers of this bug have been associated with constipation.
● Escherichia coli_D	6.561%	<i>E. coli</i> are commonly found in the gut, especially in babies. Most strains are harmless and they can be even beneficial as they help train the immune system in little ones. The <i>E. coli</i> that we detect in the gut are often different from the pathogenic <i>E. coli</i> that cause food poisoning. Pathogenic <i>E. coli</i> are spread through contaminated food and can cause stomach cramps, severe diarrhea, and sometimes kidney damage. If these symptoms are not present, there's no need to worry about low or typical levels of <i>E. coli</i>. That said, high levels of <i>E. coli</i> (even when non pathogenic) may irritate the gut, especially if beneficial microbes are missing. High numbers have been associated with eczema in children and gestational diabetes in pregnant women.
● Enterocloster clostridioformis	3.833%	<i>E. clostridioformis</i> is a variable species of bacteria that lives in your gut. In low numbers, it's not a cause for concern in healthy people. But if its numbers go up and your gut barrier is compromised, it may cause intra-abdominal infections and abscesses. <i>E. clostridioformis</i> levels may temporarily increase after antibiotic treatment because some strains are antibiotic-resistant. A good way to keep it at bay is to boost beneficial bacteria with a healthy diet and probiotics.
● Escherichia flexneri	3.610%	<i>E. flexneri</i> (previously known as <i>Shigella flexneri</i>) are unfriendly bacteria that can cause severe infection. 10,000 to 15,000 cases of <i>E. flexneri</i> infection are reported each year in the United States. Though rare, you can come in contact with <i>E. flexneri</i> after touching poop or other surfaces with the bacteria on it. You can have an infection and feel fine, but typically <i>E. flexneri</i> cause unpleasant symptoms: • Bloody diarrhea • Stomach cramps • Fever • Vomiting Since the bacteria are most common in young children, make sure to clean up thoroughly after a diaper change. Wash those hands!

● Bacteroides fragilis	3.500%	<i>B. fragilis</i> is a variable bacteria species in your gut microbiome. In low counts, it can benefit your health and well-being. However, if your <i>B. fragilis</i> levels are too high, you can get into trouble. Not only can they contribute to inflammation by damaging your gut's structure, but the species can cause serious infections. In children, high levels of <i>B. fragilis</i> may contribute to constipation. One study suggests that high levels of <i>B. fragilis</i> can also increase the risk of asthma and eczema development. It's important to keep levels of "friend or foe" in the sweet spot. <i>B. fragilis'</i> competitors in your gut community love a high-fiber diet. Stick to fruits, veggies, and complex carbohydrates.
● Citrobacter portucalensis	3.232%	<i>C. portucalensis</i> can be part of a healthy gut. However, these bacteria are often resistant to multiple antibiotics. One study found that in comparison to healthy patients, patients with colon cancer had higher levels of <i>C. portucalensis</i>. But we don't know for sure if these bacteria have an active role in this disease. On rare occasions, <i>Citrobacter</i> species may cause infection outside of the gut. This may be a concern for preterm babies, immunocompromised, elderly or debilitated patients.
● Clostridium_AQ innocuum	3.157%	<i>C. innocuum</i> is a variable species of bacteria that can live in your gut. In low numbers, it's not a cause for concern in healthy people. But because <i>C. innocuum</i> is usually resistant to the antibiotic vancomycin it may cause diarrhea after antibiotic treatment, similar to <i>C. difficile</i>. Older and immunocompromised people are especially at risk. On rare occasions, it can also cause infections outside the gut.
● Citrobacter_A amalonaticus	3.099%	<i>C. amalonaticus</i> are unfriendly bacteria from the <i>Enterobacteriaceae</i> family. These bugs have been detected in the gut of healthy people, so at low levels they probably don't cause any trouble. It's possible that their levels increase temporarily after antibiotic treatment as <i>C. amalonaticus</i> can be resistant to antibiotics. Compared to healthy people, <i>C. amalonaticus</i> have been found at higher levels in those with Crohn's disease, but we don't know for sure if these bacteria have an active role in this disease. In immunocompromised and hospitalized patients, these bacteria can rarely cause infection.
● Citrobacter_A farmeri	2.536%	
● Agathobacter rectalis	2.505%	<i>A. rectalis</i> is a beneficial bacterium found in the gut of many people. It's a butyrate producer whose levels increase when babies start eating solid foods. According to one study, very low levels of <i>A. rectalis</i> may contribute to inflammatory bowel syndrome. But too much of it may be associated with obesity. So better to keep <i>A. rectalis</i> at that sweet spot. If you need to give it a boost, you can add more fiber to your diet.
● Enterocloster bolteae	2.020%	<i>E. bolteae</i> are variable bacteria that live in the gut. In low numbers, they can contribute to gut health by producing short-chain fatty acids. In children with autism and in adults with inflammatory joint diseases, <i>E. bolteae</i> levels are increased. But we don't know for sure if these bacteria have a direct role in these conditions.

● Hungatella effluvii	1.909%	<i>H. effluvii</i> is a beneficial bacterium in your gut that produces acetate and propionate, two short-chain fatty acids that contribute to gut health. Very rarely it can cause infection, but mostly in immunocompromised and old patients.
● Parabacteroides distasonis	1.729%	<i>P. distasonis</i> is a variable species found in the gut microbiome of many healthy people. The species can disrupt your health or benefit it. Scientists have linked it to a variety of health conditions, when in high counts and outside of the gut, including: • Inflammatory bowel disease • Cardiovascular disease • Autoimmune disease • Diabetes and gestational diabetes • Abscesses However, the species may play a protective role against colorectal cancer and obesity. Luckily, having <i>P. distasonis</i> in your gut isn't much of a concern unless its populations grow out of balance.
● Bacteroides uniformis	1.654%	Lots of healthy people have <i>B. uniformis</i> in their gut. This bacterium is very good at digesting a wide range of plant-based foods. It has anti-inflammatory properties that keep your gut healthy. It may also have a role in reducing binge eating and anxiety. Breastfeeding is a good way to boost <i>B. uniformis</i> levels in babies.
● Robinsoniella peoriensis	1.611%	We don't know much about <i>R. peoriensis</i>. Only one study has reported its detection in the human gut so its role in gut health is unclear. It is known to cause infections outside of the gut, usually following trauma and/or surgery. In some cases this seems to be related to a disrupted gut barrier.
● Bacteroides cellulosilyticus	1.549%	<i>Bacteroides cellulosilyticus</i> are present in the human gut where they help break down complex carbohydrates found in plant-based foods. It contributes to gut health by producing molecules such as sugars and essential fatty acids that modulate the immune system through its anti-inflammatory properties. <i>B. cellulosilyticus</i> can also be an ally for your baby. Scientists have proposed that carbohydrate metabolism - both HMOs in breastmilk and fiber with the introduction of solids - by gut bacteria during early infancy protects against the development of allergies.
● Flavonifractor plautii	1.434%	<i>F. plautii</i> is a variable species of bacteria that lives in your gut. It can digest flavonoids, compounds found in fruits and veggies. In low numbers, it's not a cause for concern in healthy people. In fact, some studies suggest it may protect you against allergies and the flu. But if its numbers go up and your gut barrier is compromised, it may use that opportunity to cause infection.

All species

● Clostridium_Q symbiosum	1.146%	● Odoribacter splanchnicus	1.081%
● Enterocloster aldenensis	0.941%	● Hungatella sp005845265	0.650%
● Bilophila wadsworthia	0.625%	● Parabacteroides johnsonii	0.451%
● Eggerthella lenta	0.434%	● Senegalimassilia anaerobia	0.321%
● Merdimonas faecis	0.307%	● Citrobacter werkmanii	0.268%
● Phocaeicola vulgatus	0.241%	● Citrobacter freundii	0.224%
● Enterocloster lavalensis	0.194%	● Flavonifractor sp000508885	0.189%
● Mediterraneibacter sp002159505	0.177%	● Citrobacter_A sp009363175	0.167%
● Enterocloster citroniae	0.165%	● Proteus mirabilis	0.159%
● Klebsiella pneumoniae	0.150%	● Streptococcus salivarius	0.145%
● Klebsiella_A oxytoca	0.137%	● Anaerostipes sp000508985	0.127%
● Bifidobacterium animalis	0.122%	● Bifidobacterium longum	0.119%
● Streptococcus sp001556435	0.115%	● Bacteroides xylanisolvens	0.103%
● Escherichia coli	0.103%	● Intestinibacter bartlettii	0.097%
● Escherichia dysenteriae	0.096%	● Schaedlerella sp004556565	0.084%
● Muricomes contortus_B	0.082%	● Enterobacter hormaechei_A	0.080%
● Bifidobacterium dentium	0.079%	● Bifidobacterium breve	0.072%
● Enterococcus_D casseliflavus	0.069%	● Erysipelatoclostridium spiroforme	0.063%
● Alloscardovia omnicolens	0.057%	● Citrobacter_A amalonaticus_C	0.057%
● Faecalibacillus intestinalis	0.052%	● UMGS1071 sp900542375	0.049%
● Prevotellamassilia timonensis	0.048%	● Agathobacter sp900546625	0.041%
● Bacteroides fragilis_A	0.040%	● Pauljensenia turicensis	0.040%
● Prevotella bivia	0.040%	● Terrisporobacter othiniensis	0.039%
● Prevotella copri	0.038%	● Senegalimassilia sp900550055	0.037%
● Amulumruptor caecigallinarius	0.033%	● Streptococcus parasanguinis_B	0.033%
● Veillonella parvula_A	0.033%	● Gordonibacter urolithinfaciens	0.031%
● Escherichia coli_C	0.030%	● Agathobacter sp900317585	0.029%
● Bacteroides sp900552405	0.029%	● Erysipelatoclostridium ramosum	0.028%
● Parabacteroides merdae	0.028%	● Pauljensenia sp902373545	0.027%

● Bacteroides clarus	0.026%	● BX12 sp014333425	0.026%
● Streptococcus parasanguinis	0.026%	● Streptococcus anginosus_C	0.025%
● Dialister invisus	0.024%	● Chitinophaga sp003412465	0.023%
● Blautia_A sp900066505	0.022%	● Escherichia albertii	0.020%
● Phocaeicola plebeius	0.020%	● Levyella massiliensis	0.019%
● Saccharomyces cerevisiae	0.019%	● Coprobacter sp900555475	0.018%
● Fusicatenibacter saccharivorans	0.018%	● Blautia_A wexlerae_A	0.017%
● Escherichia fergusonii	0.017%	● Finegoldia magna_H	0.017%
● Phocaeicola sp900553715	0.017%	● Streptococcus anginosus	0.016%
● Butyrivibrio_A crossotus	0.015%	● Phocaeicola massiliensis	0.015%
● Proteus mirabilis_B	0.014%	● Alistipes_A ihumii	0.013%
● Bacteroides caecimuris	0.013%	● CAG-83 sp004557655	0.013%
● Duncaniella dubosii	0.013%	● Dysosmobacter sp014297375	0.013%
● Prevotella disiens	0.013%	● Actinomyces oris	0.012%
● Phocaeicola sartorii	0.012%	● Bacteroides sp900765805	0.011%
● Escherichia marmotae	0.011%	● Peptostreptococcus anaerobius	0.011%
● RC9 sp009774765	0.011%	● Bacteroides faecis	0.010%
● Bacteroides sp003463205	0.010%	● Blautia coccoides	0.010%
● Citrobacter youngae	0.010%	● Parabacteroides sp900541965	0.010%

Our high confidence detection limit is 0.01% abundance. Very low abundance organisms can be important members of the microbiome but also very difficult to accurately detect. Detections below 0.01% abundance have an increased risk of being false positives. However, in our testing, we have observed that most false positive identifications are closely related to a detected species.

● Citrobacter braakii	0.009%	● Porphyromonas asaccharolytica	0.009%
● Prevotella stercorea	0.009%	● Actinomyces naeslundii	0.008%
● Bacteroides acidifaciens	0.008%	● Caecibacter massiliensis	0.008%
● CAG-81 sp900066785	0.008%	● Dialister sp900555245	0.008%
● Massilimaliae timonensis	0.008%	● Streptococcus parasanguinis_D	0.008%
● Xuanwuvirus xv520873	0.008%	● Anaerococcus vaginalis_B	0.007%
● Bacteroides sp900066265	0.007%	● Intestinibacter sp900553485	0.007%
● Mobiluncus curtisii	0.007%	● Peduovirus P2	0.007%
● Phocaeicola sp900546095	0.007%	● Sellimonas sp002161525	0.007%

● Wadgaonvirus wv5004651	0.007%	● Afonbuvirus faecalis	0.006%
● Atopobium minutum	0.006%	● Bacteroides finegoldii	0.006%
● Blautia_A obeum_B	0.006%	● Peduovirus magyaro	0.006%
● Peptoniphilus_A harei_A	0.006%	● Peptoniphilus_C sp900343085	0.006%
● Phocaeicola sp002493165	0.006%	● RC9 sp900546925	0.006%
● Robinsoniella sp900539655	0.006%	● Ruthenibacterium lactatiformans	0.006%
● Veillonella dispar_A	0.006%		

References

- [1] A. Uzan-Yulzari et al., "Neonatal antibiotic exposure impairs child growth during the first six years of life by perturbing intestinal microbial colonization," *Nat Commun*, vol. 12, no. 1, p. 443, 2021, doi: 10.1038/s41467-020-20495-4.
- [2] K. E. Fujimura et al., "Neonatal gut microbiota associates with childhood multisensitized atopy and T cell differentiation," *Nat Med*, vol. 22, no. 10, pp. 1187–1191, 2016, doi: 10.1038/nm.4176.
- [3] M. B. Azad et al., "Infant gut microbiota and food sensitization: associations in the first year of life," *Clin Exp Allergy*, vol. 45, no. 3, pp. 632–643, 2015, doi: 10.1111/cea.12487.
- [4] H. M. Tun et al., "Roles of Birth Mode and Infant Gut Microbiota in Intergenerational Transmission of Overweight and Obesity From Mother to Offspring," *Jama Pediatr*, vol. 172, no. 4, p. 368, 2018, doi: 10.1001/jamapediatrics.2017.5535.
- [5] T. Vatanen et al., "The human gut microbiome in early-onset type 1 diabetes from the TEDDY study," *Nature*, vol. 562, no. 7728, pp. 589–594, 2018, doi: 10.1038/s41586-018-0620-2.
- [6] C. J. Stewart et al., "Temporal development of the gut microbiome in early childhood from the TEDDY study," *Nature*, vol. 562, no. 7728, pp. 583–588, 2018, doi: 10.1038/s41586-018-0617-x.
- [7] S. Tamburini, N. Shen, H. C. Wu, and J. C. Clemente, "The microbiome in early life: implications for health outcomes," *Nat Med*, vol. 22, no. 7, pp. 713–722, 2016, doi: 10.1038/nm.4142.
- [8] M. J. Ege, "The Hygiene Hypothesis in the Age of the Microbiome," *Ann Am Thorac Soc*, vol. 14, no. Supplement_5, pp. S348–S353, 2017, doi: 10.1513/annalsats.201702-139aw.
- [9] B. M. Henrick et al., "Bifidobacteria-mediated immune system imprinting early in life," *Cell*, 2021, doi: 10.1016/j.cell.2021.05.030.
- [10] A. Marcobal et al., "Bacteroides in the Infant Gut Consume Milk Oligosaccharides via Mucus-Utilization Pathways," *Cell Host Microbe*, vol. 10, no. 5, pp. 507–514, 2011, doi: 10.1016/j.chom.2011.10.007.
- [11] T. Wilmski et al., "Gut microbiome pattern reflects healthy ageing and predicts survival in humans," *Nat Metabolism*, vol. 3, no. 2, pp. 274–286, 2021, doi: 10.1038/s42255-021-00348-0.
- [12] M. A. Jackson et al., "Signatures of early frailty in the gut microbiota," *Genome Med*, vol. 8, no. 1, p. 8, 2016, doi: 10.1186/s13073-016-0262-7.
- [13] J. M. Pickard, M. Y. Zeng, R. Caruso, and G. Núñez, "Gut microbiota: Role in pathogen colonization, immune responses, and inflammatory disease," *Immunol Rev*, vol. 279, no. 1, pp. 70–89, 2017, doi: 10.1111/imr.12567.
- [14] O. Manor et al., "Health and disease markers correlate with gut microbiome composition across thousands of people," *Nat Commun*, vol. 11, no. 1, p. 5206, 2020, doi: 10.1038/s41467-020-18871-1.
- [15] M. Scudellari, "Cleaning up the hygiene hypothesis," *Proc National Acad Sci*, vol. 114, no. 7, pp. 1433–1436, 2017, doi: 10.1073/pnas.1700688114.
- [16] R. Debray, R. A. Herbert, A. L. Jaffe, A. Crits-Christoph, M. E. Power, and B. Koskella, "Priority effects in microbiome assembly," *Nat Rev Microbiol*, vol. 20, no. 2, pp. 109–121, 2022, doi: 10.1038/s41579-021-00604-w.
- [17] R. Debray, R. A. Herbert, A. L. Jaffe, A. Crits-Christoph, M. E. Power, and B. Koskella, "Priority effects in microbiome assembly," *Nat Rev Microbiol*, vol. 20, no. 2, pp. 109–121, 2022, doi: 10.1038/s41579-021-00604-w.
- [18] C. Huttenhower et al., "Structure, function and diversity of the healthy human microbiome," *Nature*, vol. 486, no. 7402, pp. 207–214, 2012, doi: 10.1038/nature11234.
- [19] P. I. Costea et al., "Enterotypes in the landscape of gut microbial community composition," *Nat Microbiol*, vol. 3, no. 1, pp. 8–16, 2018, doi: 10.1038/s41564-017-0072-8.
- [20] W. E. Anthony et al., "Acute and persistent effects of commonly used antibiotics on the gut microbiome and resistome in healthy adults," *Cell Reports*, vol. 39, no. 2, p. 110649, 2022, doi: 10.1016/j.celrep.2022.110649.
- [21] A. Palleja et al., "Recovery of gut microbiota of healthy adults following antibiotic exposure," *Nat Microbiol*, vol. 3, no. 11, pp. 1255–1265, 2018, doi: 10.1038/s41564-018-0257-9.
- [22] A. Bell et al., "Elucidation of a sialic acid metabolism pathway in mucus-foraging *Ruminococcus gnavus* unravels mechanisms of bacterial adaptation to the gut," *Nat Microbiol*, vol. 4, no. 12, pp. 2393–2404, 2019, doi: 10.1038/s41564-019-0590-7.
- [23] M. T. Henke, D. J. Kenny, C. D. Cassilly, H. Vlamakis, R. J. Xavier, and J. Clardy, "Ruminococcus gnavus, a member of the human gut microbiome associated with Crohn's disease, produces an inflammatory polysaccharide," *Proc National Acad Sci*, vol. 116, no. 26, pp. 12672–12677, 2019, doi: 10.1073/pnas.1904099116.
- [24] W. Li et al., "Ecological and network analyses identify four microbial species with potential significance for the diagnosis/treatment of ulcerative colitis (UC)," *Bmc Microbiol*, vol. 21, no. 1, p. 138, 2021, doi: 10.1186/s12866-021-02201-6.
- [25] F. Blachier, M. Beaumont, and E. Kim, "Cysteine-derived hydrogen sulfide and gut health," *Curr Opin Clin Nutr*, vol. 22, no. 1, pp. 68–75, 2019, doi: 10.1097/mco.0000000000000526.
- [26] S. C. Peck, K. Denger, A. Burrichter, S. M. Irwin, E. P. Balskus, and D. Schleheck, "A glycol radical enzyme enables hydrogen sulfide production by the human intestinal bacterium *Bifidobacterium wadsworthia*," *Proc National Acad Sci*, vol. 116, no. 8, pp. 3171–3176, 2019, doi: 10.1073/pnas.1815661116.
- [27] H. M. HAMER, D. JONKERS, K. VENEMA, S. VANHOUTVIN, F. J. TROOST, and R. -J. BRUMMER, "Review article: the role of butyrate on colonic function," *Aliment Pharm Therap*, vol. 27, no. 2, pp. 104–119, 2008, doi: 10.1111/j.1365-2036.2007.03562.x.
- [28] Y. Furusawa et al., "Commensal microbe-derived butyrate induces the differentiation of colonic regulatory T cells," *Nature*, vol. 504, no. 7480, pp. 446–450, 2013, doi: 10.1038/nature12721.
- [29] H. Sokol et al., "Faecalibacterium prausnitzii is an anti-inflammatory commensal bacterium identified by gut microbiota analysis of Crohn disease patients," *Proc National Acad Sci*, vol. 105, no. 43, pp. 16731–16736, 2008, doi: 10.1073/pnas.0804812105.
- [30] D. Ríos-Covián, P. Ruas-Madiedo, A. Margolles, M. Gueimonde, C. G. de los Reyes-Gavilán, and N. Salazar, "Intestinal Short Chain Fatty Acids and their Link with Diet and Human Health," *Front Microbiol*, vol. 7, p. 185, 2016, doi: 10.3389/fmicb.2016.00185.
- [31] Q. Zhou et al., "Association Between Gut Akkermansia and Metabolic Syndrome is Dose-Dependent and Affected by Microbial Interactions: A Cross-Sectional Study," *Diabetes Metabolic Syndrome Obes Targets Ther*, vol. 14, pp. 2177–2188, 2021, doi: 10.2147/dmso.s311388.